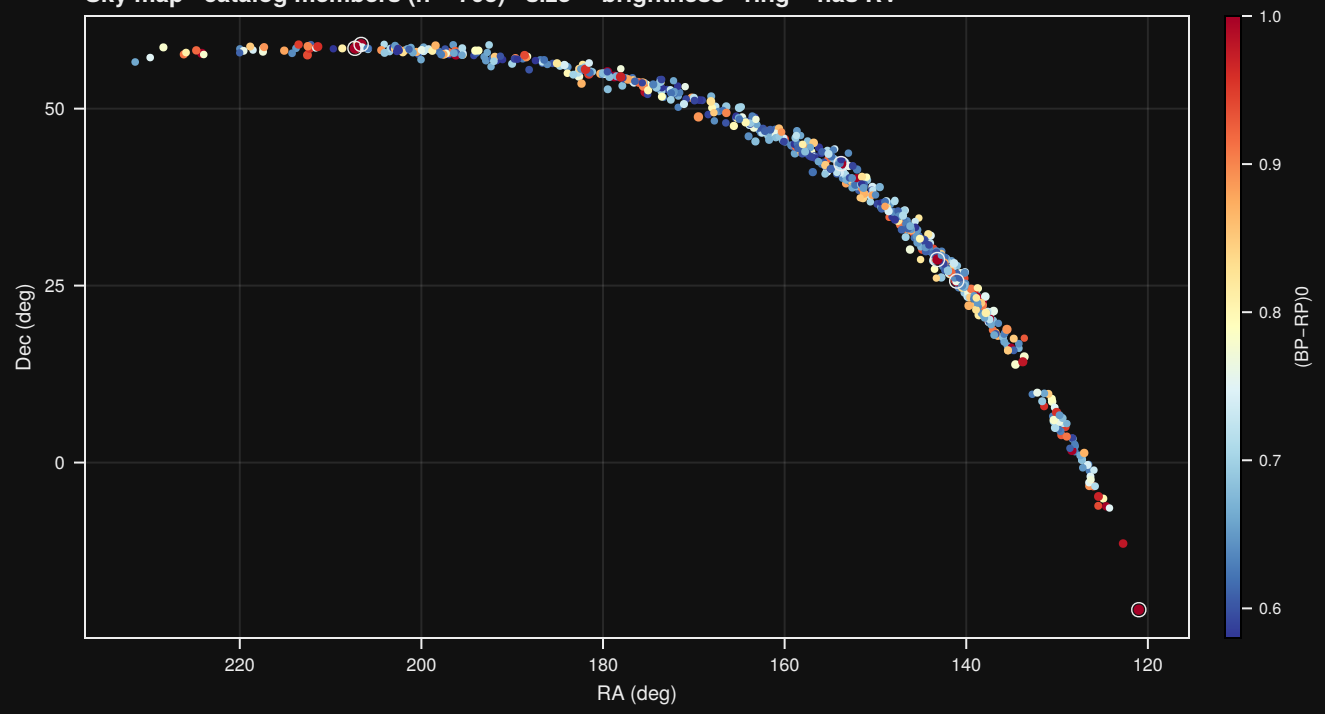
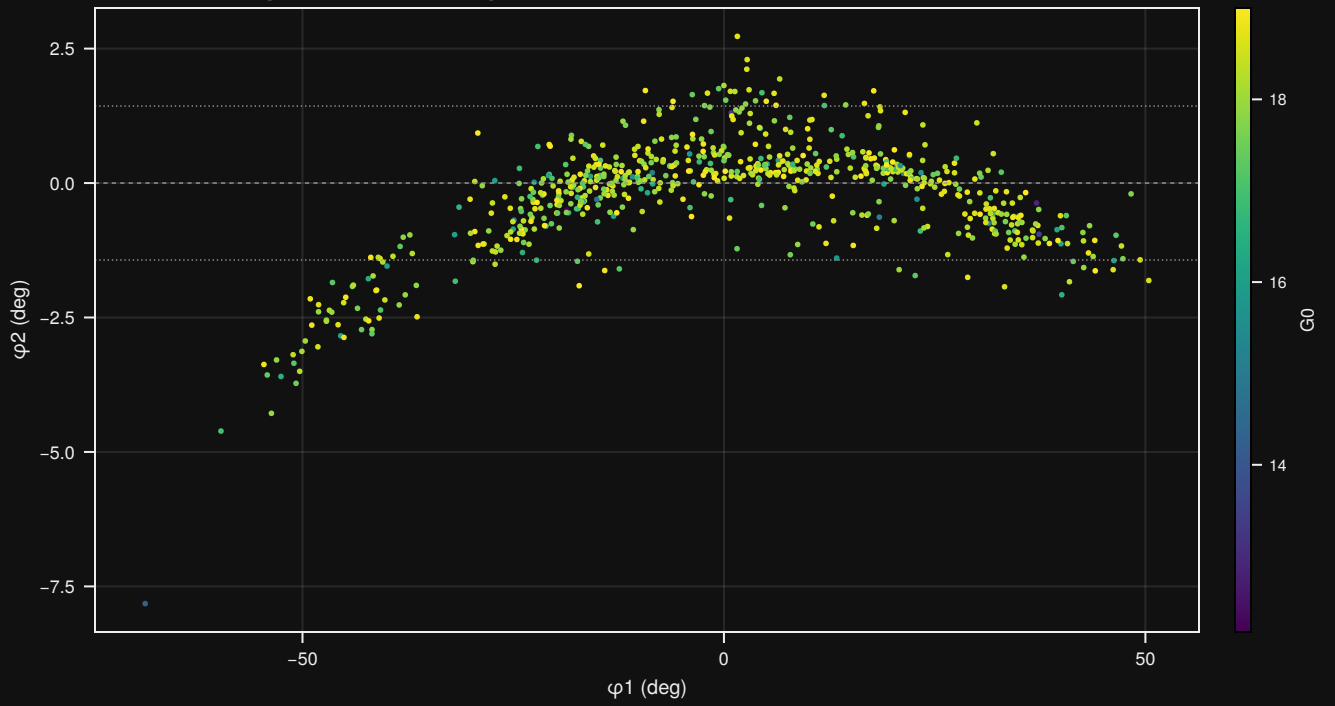
GD-1

MEMBERSHIP	
Stars in catalog	703
with Gaia GSP-Phot	703 (100%)
with radial velocity	6
Stream id (sID)	53
GEOMETRY · great-circle fit	
Centroid RA, Dec	159.3, 41.0 deg
Centroid l, b	173.6, 54.2 deg
Pole RA, Dec	34.1, 30.3 deg
Length (φ1, 2–98%)	91.5 deg
Width σ(φ2), robust	0.72 deg
KINEMATICS	
pmRA* med ± σ	-5.99 ± 2.10 mas/yr
pmDE med ± σ	-11.28 ± 2.42 mas/yr
μφ1* med ± σ	-12.27 ± 1.57 mas/yr
μφ2 med ± σ	-3.25 ± 0.73 mas/yr
Parallax (cat.) med	0.127 ± 0.162 mas
Parallax (wtd, ZP-corr.)	0.133 ± 0.010 mas
Distance	d = 7.5 ± 0.6 kpc
Physical length	12.03 kpc
RV median ± σ(MAD)	-4.4 ± 261.3 km/s
RV gradient dV/dφ1	-6.01 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.2–19.0 (18.2)
(BP−RP)0 median	0.68
GSP-Phot [M/H] (n=703)	-1.97 ± 0.78
GSP-Phot Teff median	6091 K
RUWE > 1.4	0.0 %
PM clusters (largest)	36 (54 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	703 | 1468
n_RV catalog | paper	6 | 323
<[Fe/H]> paper	-2.49
Stellar mass (paper)	14000 Msun
Pole offset vs paper	0.9 deg
LABEL CHECK	n differs from paper!

Sky map · catalog members (n = 703) · size ∝ brightness · ring = has RV

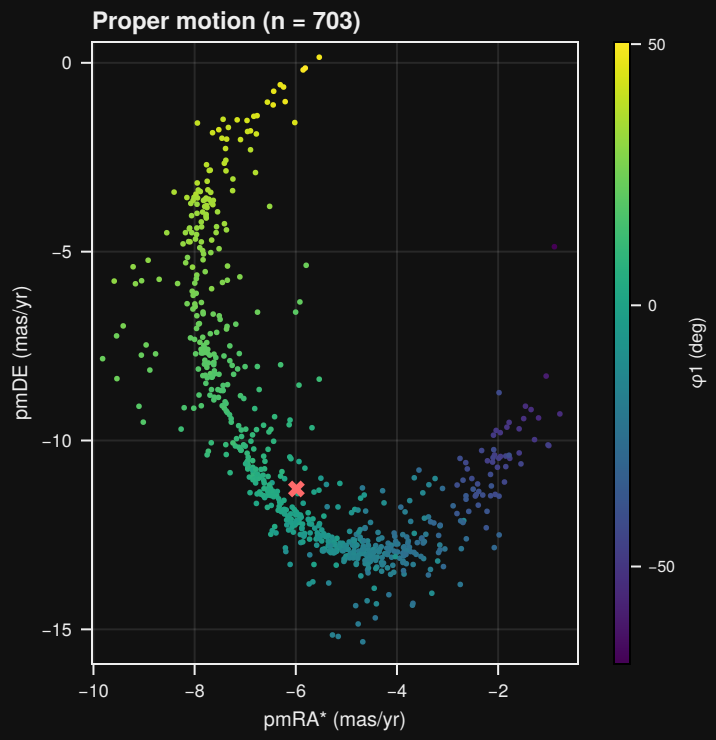


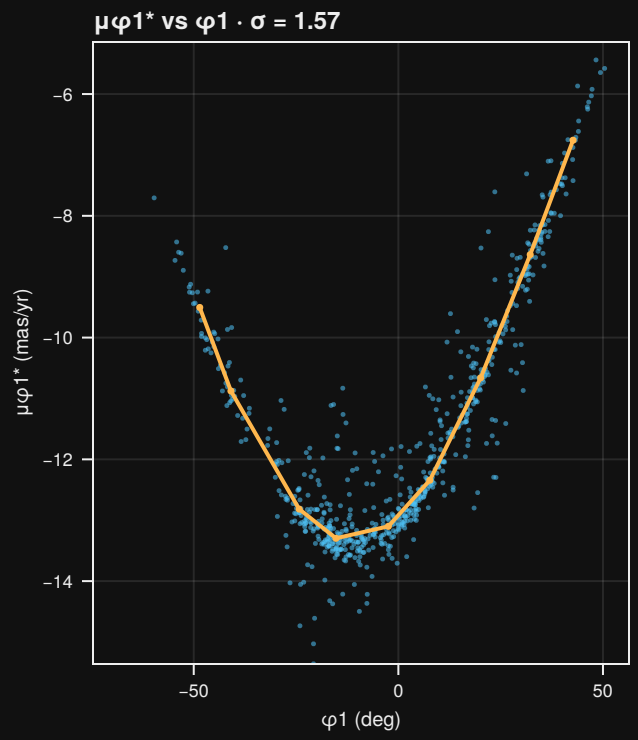
Stream frame (great-circle fit) · length 91.5° · width σ 0.72°

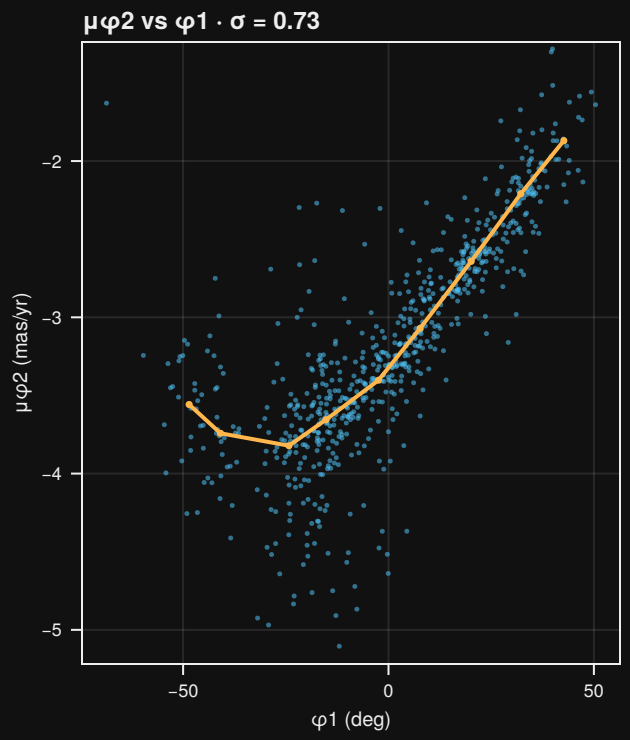


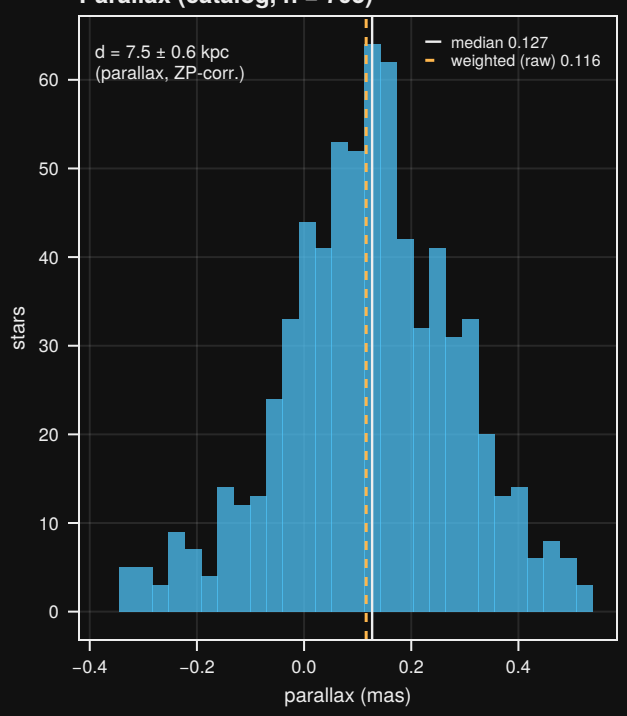
GD-1

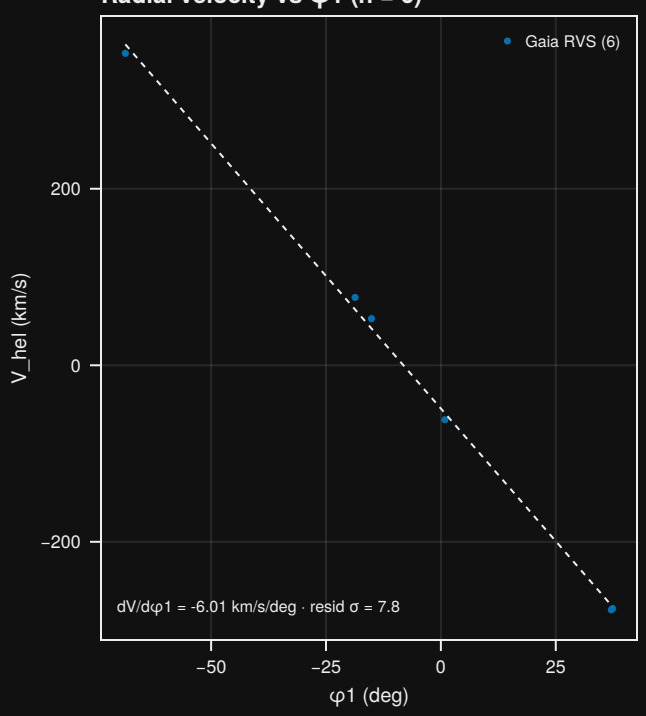
Kinematics · proper motion · parallax · radial velocity

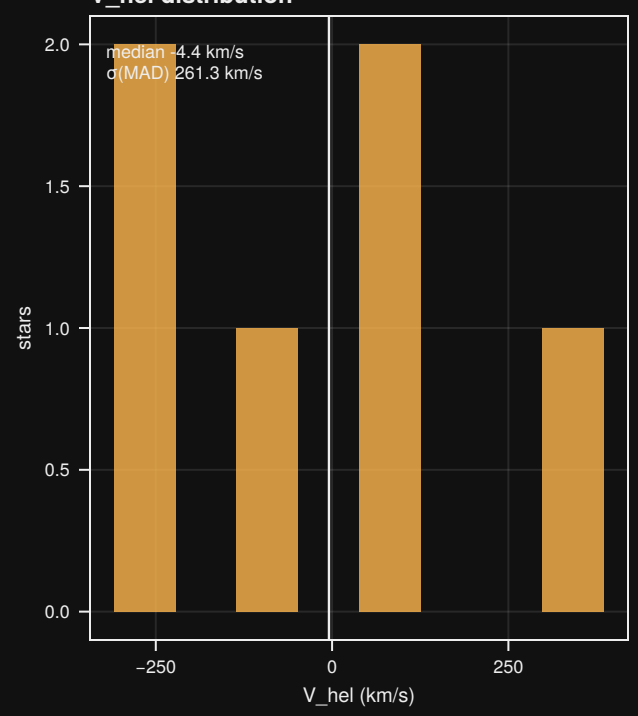






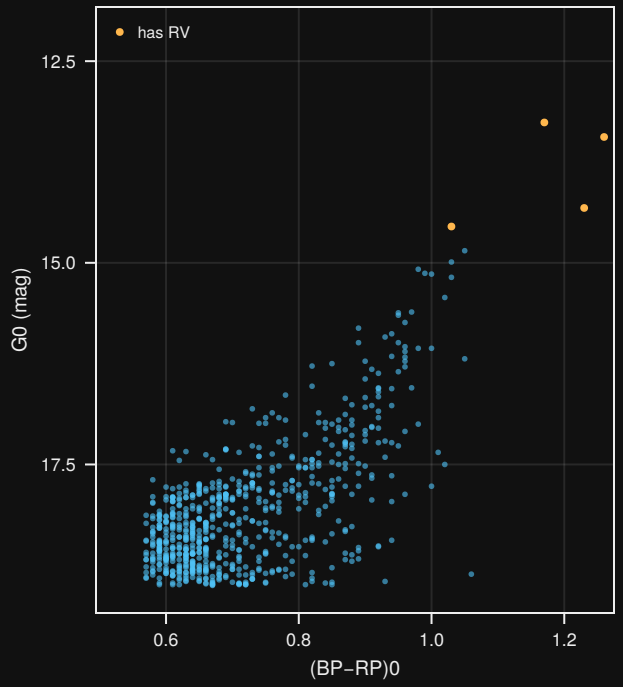




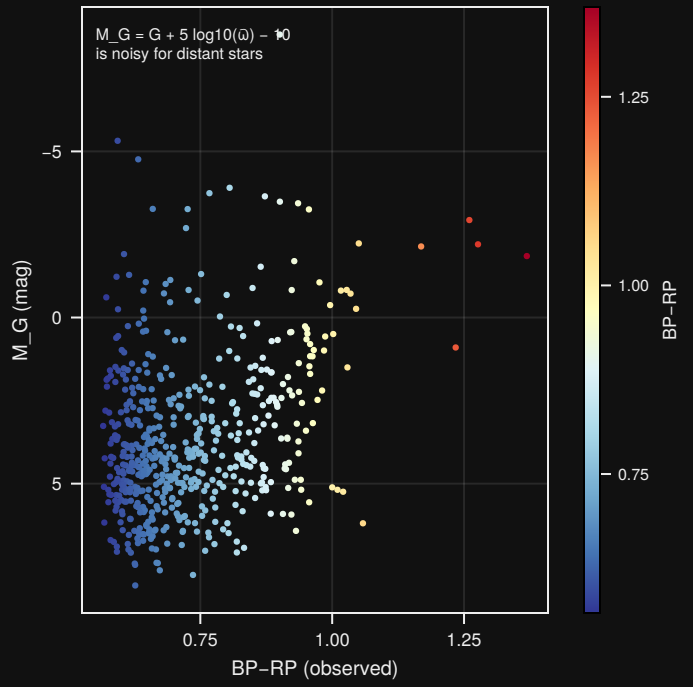


GD-1

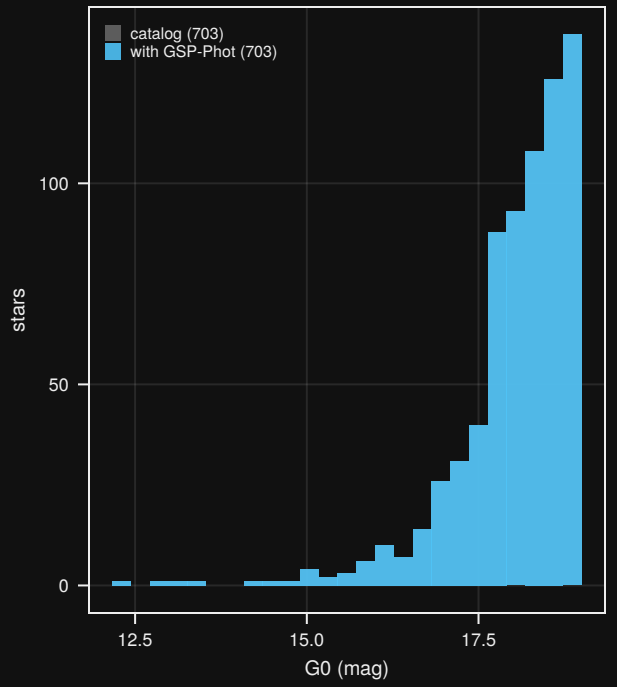
CMD · dereddened (n = 703)



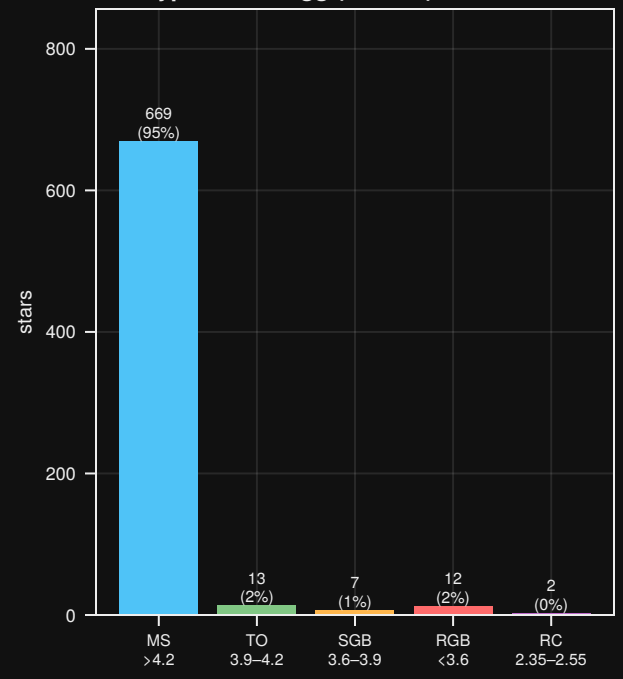
HR diagram · ϖ>0 (n = 555/703)



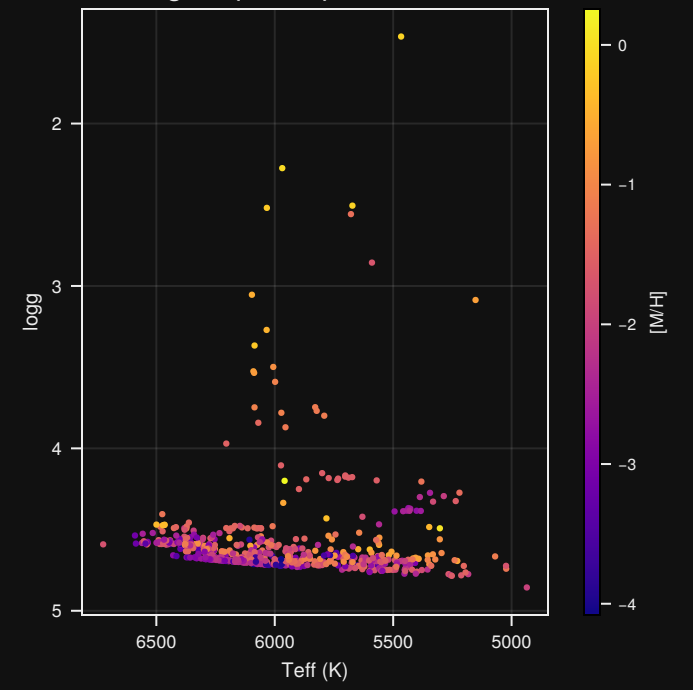
G0 & GSP-Phot coverage



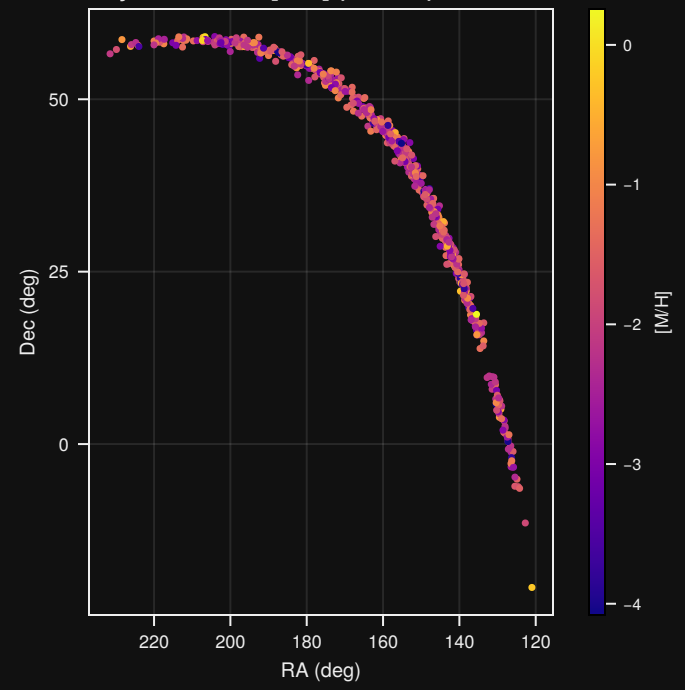
Star types from logg (n = 703)



Kiel diagram (n = 703)



Sky · GSP-Phot [M/H] (n = 703)

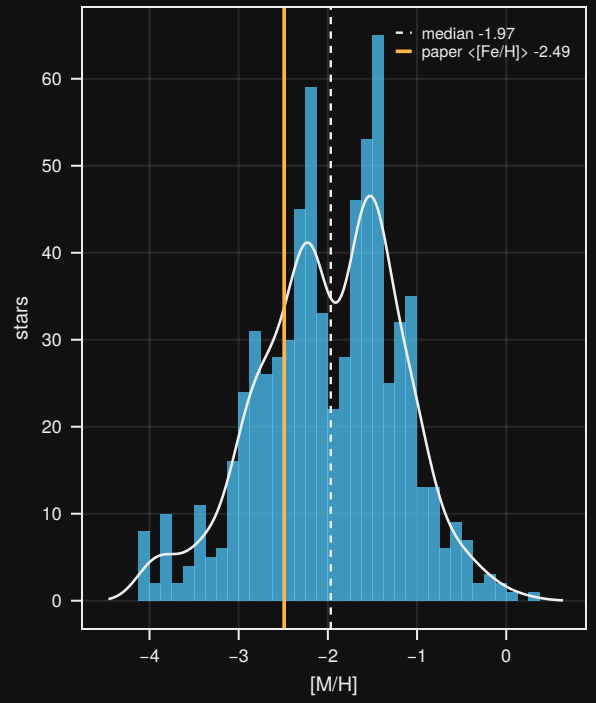


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

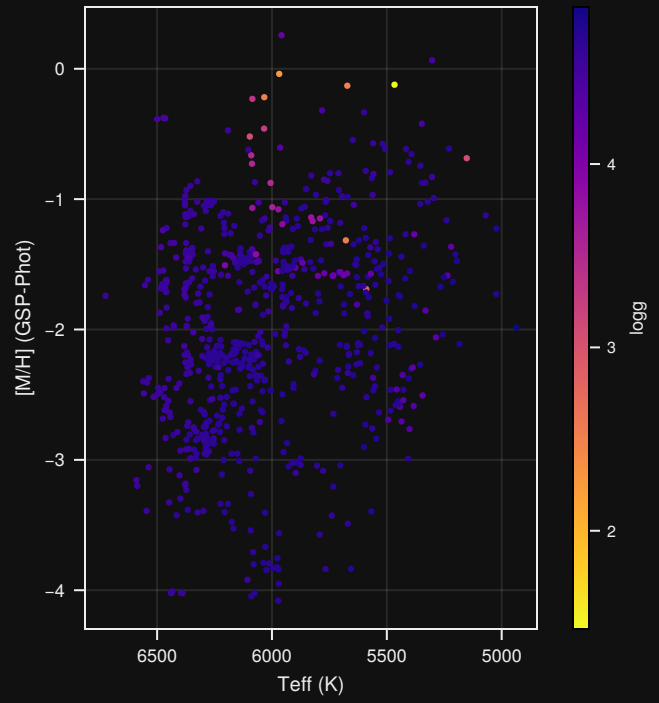
GD-1

Metallicity · age · astrometric quality

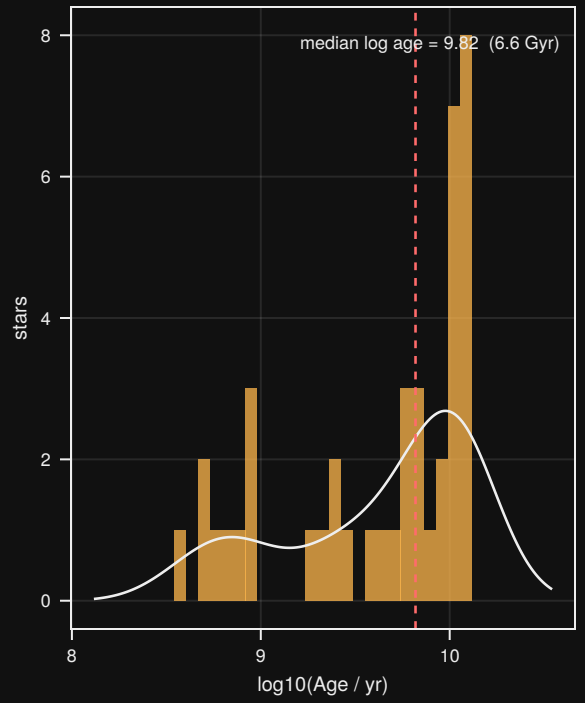
[M/H] GSP-Phot (n = 703)



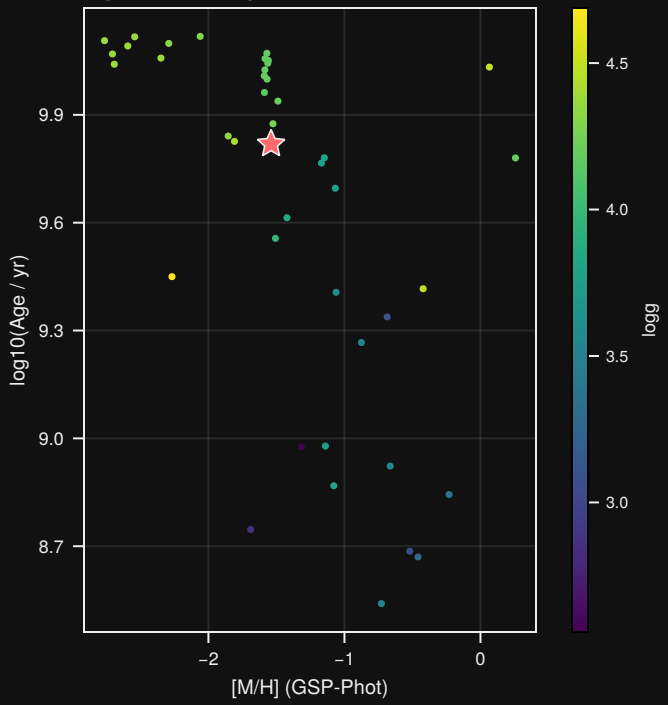
Teff vs [M/H] (n = 703)



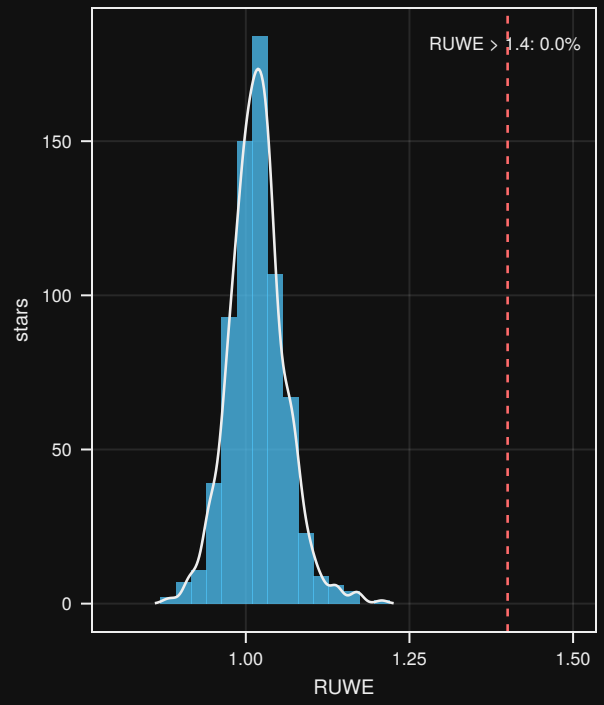
FLAME age (n = 42)



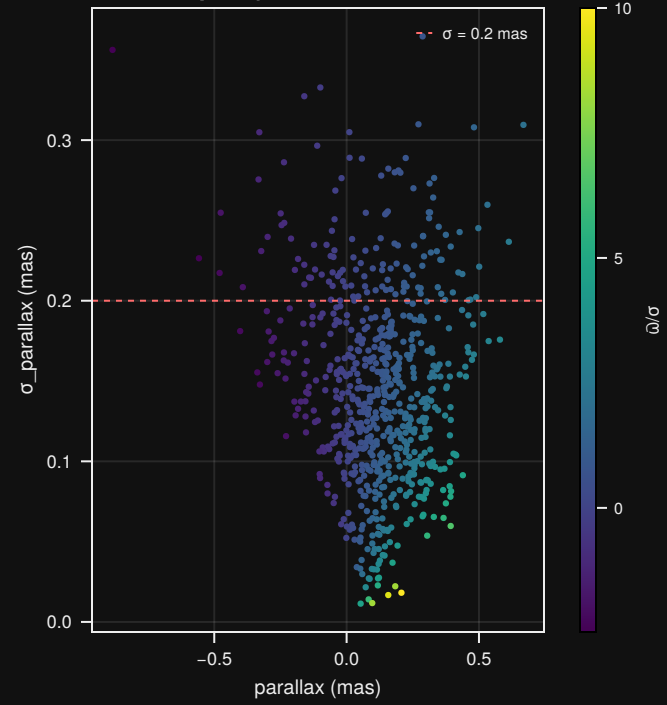
Age–metallicity (n = 42)



RUWE (n = 703)

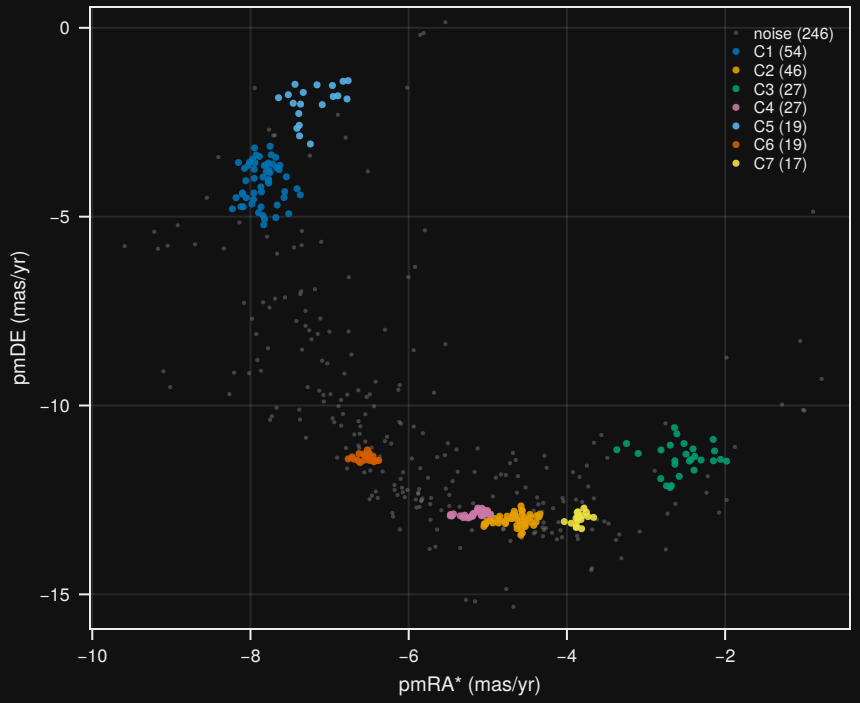


Parallax quality (n = 703)

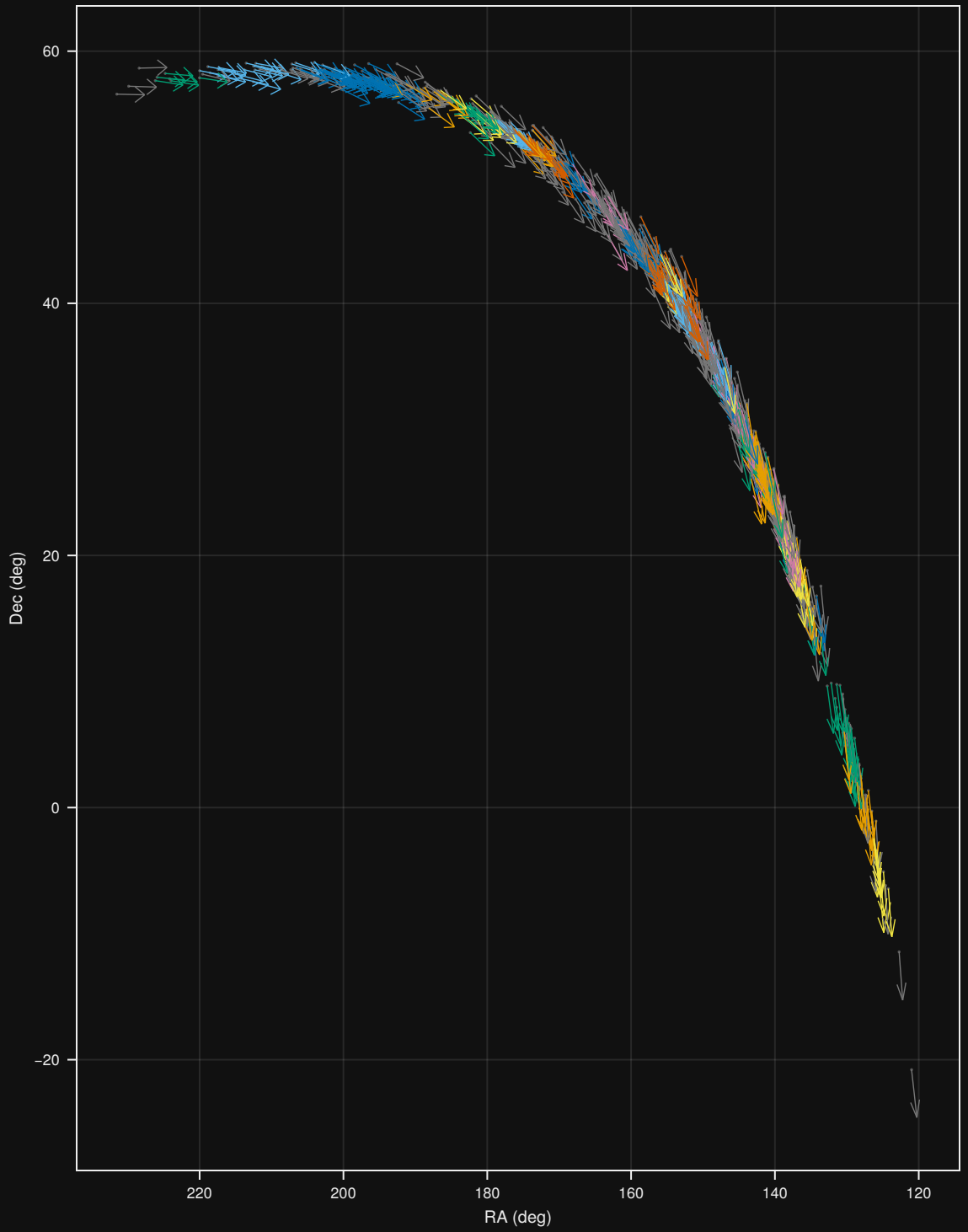


GD-1

PM clustering · HDBSCAN · 36 cluster(s) · noise 35%



Proper-motion directions (arrows) coloured by cluster



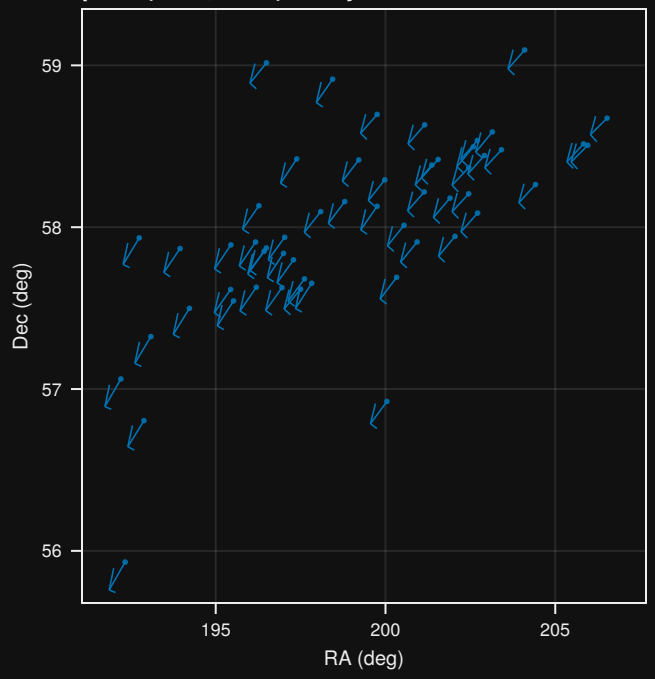
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	246	-5.95	-11.15	1.77	-1.66	3
C1	54	-7.84	-4.03	0.18	-2.13	0
C2	46	-4.63	-13.05	0.15	-2.21	1
C3	27	-2.58	-11.43	0.26	-2.11	0
C4	27	-5.13	-12.88	0.15	-1.90	0
C5	19	-7.33	-1.85	0.26	-1.41	1
C6	19	-6.55	-11.41	0.11	-2.00	0
C7	17	-3.86	-12.96	0.05	-1.69	0
C8	17	-6.82	-10.96	0.06	-1.99	0

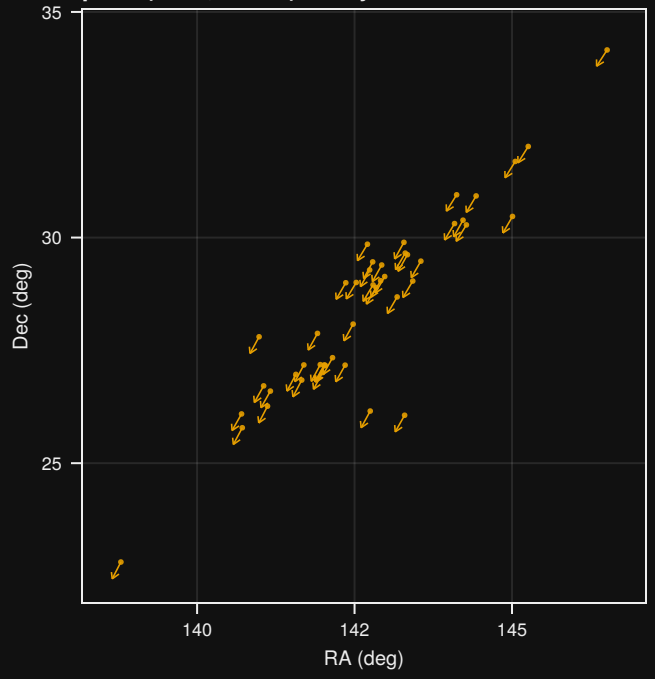
GD-1

Per-cluster proper-motion vector maps

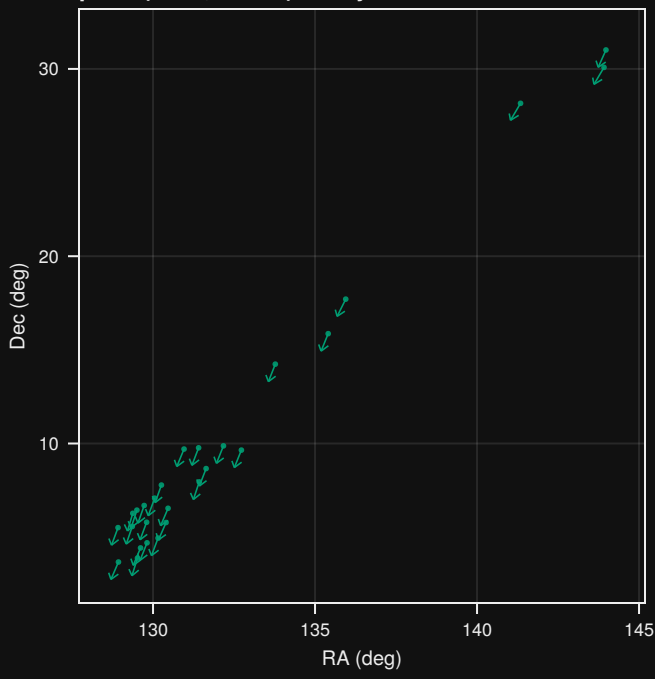
Cluster C1 · n = 54
pm = (-7.84, -4.03) mas/yr



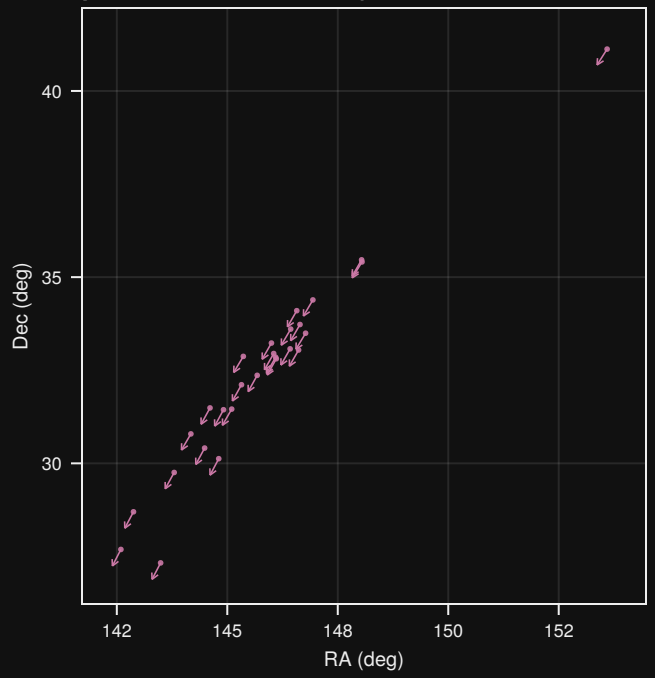
Cluster C2 · n = 46
pm = (-4.63, -13.05) mas/yr



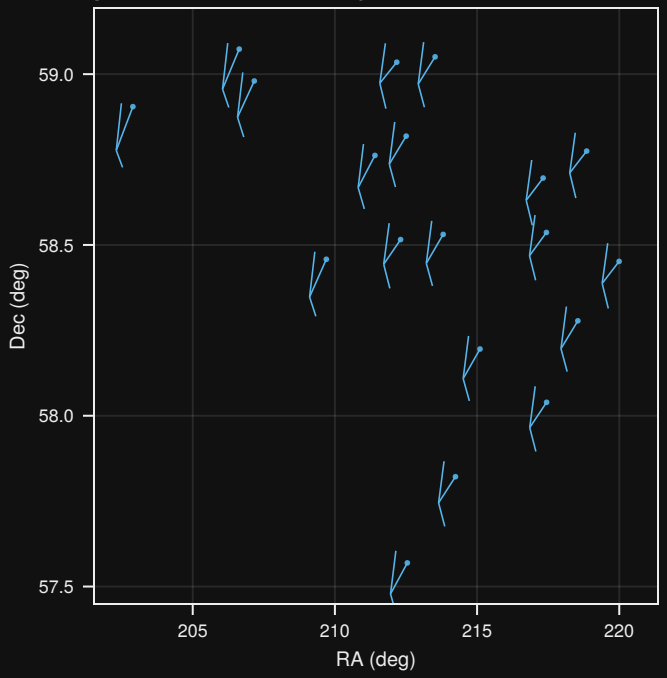
Cluster C3 · n = 27
pm = (-2.58, -11.43) mas/yr



Cluster C4 · n = 27
pm = (-5.13, -12.88) mas/yr



Cluster C5 · n = 19
pm = (-7.33, -1.85) mas/yr



Noise · n = 246

